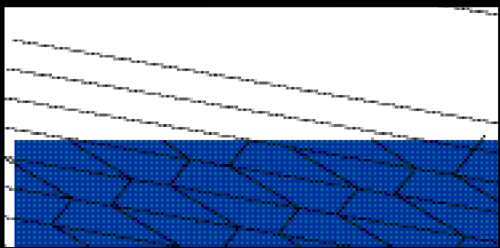
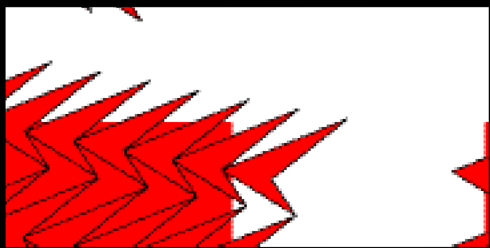
4 SIDES

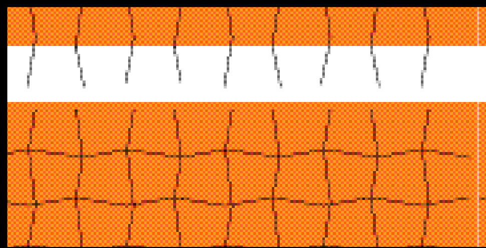
... are called QUADRILATERALS

Shapes that fit

Shapes that fit together like tiles, without any gaps, are said to ***tesselate***. The pictures below show that identical quadrilaterals always tesselate, whatever their shape. Triangles and hexagons also tesselate, but other polygons don't. So why do some shapes but not others tesselate? It all depends on the angles in the corners. If you can fit the corners together to make a full circle (360°) or a half circle (180°), the shapes will tesselate.



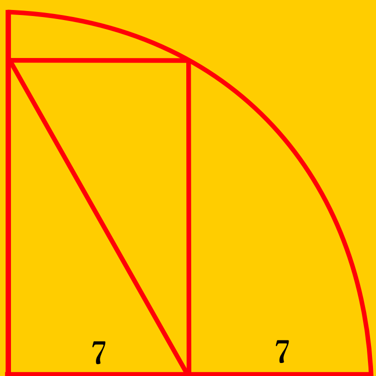




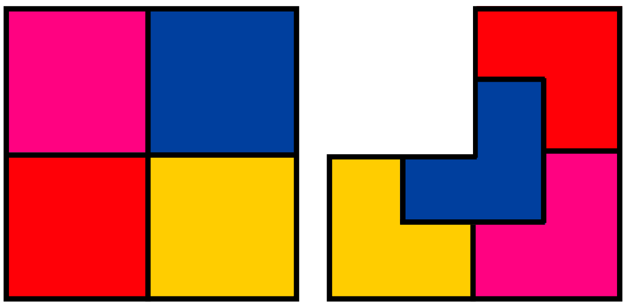
Do it yourself

You can prove that four-sided shapes always tesselate. Here's how. Use a ruler to draw ***any*** four-sided shape on a stack of about 12 pieces of newspaper. Cut out all 12 pieces at once (ask an adult to help if necessary). Use the cutouts to make a tiling pattern. You might find it tricky at first, so here's a hint: *start by lining up matching sides, but with the neighbouring shapes pointing in opposite directions.*

3 Can you work out the length of the diagonal line in the picture below?



4 The square below has been divided into four identical pieces, and the L-shaped figure has also been dissected into four identical pieces. Can you dissect a square into ***five identical pieces*** (of some shape)?



5 A plane needs to fly 140 km from A to B, but there's a 50 kph wind blowing northeast. To allow for the wind, the pilot steers towards C, which is 100 km away. If he heads towards C at 200 kph, when will he arrive at B?